09 · Case 6, part 1: what insulin does

The words of this track, as it says them. Bold brown is the examiner.

Exam case 6. Diabetes.

Larry has recently been diagnosed with type 2 diabetes mellitus, an endocrine disorder, after what seems to be many years of uncontrolled hyperglycaemia. He must measure his blood sugar every day as he works with his diabetes nurse, and he'll need insulin injections and other medications. Larry is a little overwhelmed, and isn't sure what this insulin is for. As his nurse, you need a robust understanding of what insulin does. In type 1 diabetes the body fails to make insulin. In type 2, the body can't respond to insulin when it's secreted.

Case 6 is two tracks. This one is insulin itself.

Two marks. Where in the body, specifically, is insulin made?

In the endocrine pancreas, in the islets of Langerhans. Specifically, the beta cells of the islets make insulin, which lowers blood sugar: after a meal, it tells the body to store the extra glucose. The alpha cells make glucagon. Her memory trick: glucagon goes with the alpha cells. Think of the A in both.

Three marks. What hormone, or hormones, act in opposition to insulin, and where are they made?

The hook: glucagon first, then the stress pair, adrenaline and cortisol, then growth and thyroid.

Glucagon, from the alpha cells of the pancreas, which raises blood glucose. Adrenaline, also called epinephrine, from the adrenal medulla, and cortisol, from the adrenal cortex. Both of those run the stress response, fight or flight and long-term stress, when the body needs ready access to glucose. And growth hormone, from the anterior pituitary, and the thyroid hormones, which are tied to metabolism. Changes in them can raise blood sugar too.

Four marks. Explain what insulin does at a cellular level. Think about how it travels in the body, what it binds to, and what happens.

The hook: travel, bind, move GLUT 4, let glucose in.

Insulin travels through the body in the blood. It binds to insulin receptors on the surface of insulin-sensitive cells. Once it's bound, a cascade of events inside the cell moves the GLUT 4 transporter from inside the cell into the cell membrane. And in the membrane, GLUT 4 is a transport protein that lets glucose move into the cell, down its concentration gradient, by facilitated diffusion.

One mark each. Insulin's role in the liver. In the muscles. And in the fat tissue.

The liver: it stimulates the conversion of glucose to glycogen, the storage form of glucose, by glycogenesis. Enzymes join the glucose molecules together into the larger glycogen polysaccharide.

The muscles: the same. Glucose to glycogen, by glycogenesis. Her formative test words it another way: insulin stimulates muscle cells to insert GLUT 4 into the membrane, to let glucose in. Both are true, so give the one the question is after.

The fat tissue: it stimulates the conversion of glucose to fat, by lipogenesis. Fat is the most efficient storage form: more energy in a smaller space.

Her formative test has three more blanks on this, and they're easy marks if the words are ready. Glycogenesis, glucose to glycogen, is stimulated by insulin. Glycogenolysis, breaking glycogen down, is stimulated by glucagon. And gluconeogenesis and glycogenolysis together are stimulated by glucagon and epinephrine.

Once more. What insulin does.

Insulin is made by the beta cells of the islets of Langerhans, in the endocrine pancreas, and it lowers blood sugar; the alpha cells make glucagon. Its opponents: glucagon, adrenaline from the adrenal medulla, cortisol from the adrenal cortex, growth hormone, and the thyroid hormones. At a cell, insulin travels in the blood, binds its receptor, and a cascade moves GLUT 4 into the membrane, letting glucose in by facilitated diffusion. In the liver and muscles it drives glycogenesis, glucose to glycogen. In fat tissue, lipogenesis.